

ANANYA SRIDHAR

Speech AI & Audio ML Engineer — ASR, TTS & Acoustic Modeling

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PROFESSIONAL SUMMARY

Speech AI and Audio Machine Learning Engineer with hands-on expertise building production ASR, TTS, and phoneme-level acoustic verification pipelines across 8+ Indian languages. Deep experience optimizing Faster-Whisper, Vosk, Coqui TTS, and Kokoro ONNX alongside digital audio signal processing (Librosa, Praat-Parselmouth). Proven track record deploying low-latency clinical speech therapy systems and fintech conversational audio pipelines with strict real-time guarantees.

TECHNICAL SKILLS

- Speech AI & ASR/TTS:** Faster-Whisper, OpenAI Whisper, Vosk, Coqui TTS, Kokoro ONNX, Sarvam AI, Wav2Vec2, Epitran (G2P), Pronunciation Scoring, Forced Alignment
- Audio DSP & Feature Extraction:** Librosa, Praat-Parselmouth, SoundFile, Spectral Noise Reduction (noisereduce), Formant (F1/F2) Tracking, Jitter/Shimmer Extraction, Pitch Contours
- AI & Agentic Systems:** LangGraph, LangChain, PyTorch, Hugging Face Transformers, ChromaDB, Dense RAG, Reinforcement Learning (Q-learning)
- Computer Vision & Systems:** OpenCV, MediaPipe Tasks, DINOv2, Python, C++, FastAPI, WebSockets, PostgreSQL, Docker, Azure Cloud, React.js, Linux

PROFESSIONAL EXPERIENCE

Manas Learning — AI/ML Engineer Intern (Speech AI & Systems)

Mar 2026 – Present | Remote / Mumbai, India

- Multilingual Speech Diagnostic Pipeline:** Architected a pediatric speech assessment engine chaining Faster-Whisper, Vosk, and Kokoro/Coqui TTS with phoneme verification across 8 Indian languages.
- Phoneme-Level Scoring Engine:** Implemented acoustic feature analysis (formant shifts, duration ratios, and G2P phonetic alignments via Librosa and Epitran) to classify articulatory errors (place, manner, voicing).
- Adaptive Practice Engine:** Developed a Q-learning difficulty controller dynamically tuning assessment difficulty based on real-time acoustic accuracy metrics.
- Real-World Impact:** Delivered sub-300ms streaming audio inference for clinical speech assessment across vaaksudhi.manaslearning.com and vaakify.manaslearning.com, enabling automated articulatory error feedback for therapists.

Nivesh AI — AI/ML Engineer Intern (Audio ML & RAG)

Jun 2026 – Present | Mumbai, India

- Fintech Speech Ingestion Pipeline:** Built production audio transcription and summarization pipelines on Azure supporting 10,000+ mutual fund distributors (MFDs).
- Multilingual STT Optimization:** Evaluated and tuned Sarvam AI and Whisper models with spectral denoising (noisereduce), improving transcription fidelity on mixed Indian English/Hindi advisor conversations.
- Automated Advisory Documentation:** Implemented a dense RAG pipeline parsing conversation transcripts into structured CRM action notes, cutting manual documentation turnaround by 80%.
- Real-World Impact:** Processed thousands of complex multilingual advisor call recordings into structured financial summaries with zero hallucination.

FEATURED PROJECTS

Vaaksudhi — Pediatric Speech & Diagnostic Multi-Agent Platform

[Live Platform](#) · [GitHub: Agentic-AI](#)

Python, LangGraph, Faster-Whisper, Vosk, Kokoro ONNX, Librosa, Epitran, Q-Learning, FastAPI

- Architected an autonomous multi-agent clinical diagnostic pipeline coordinating 4 clinical assessment modules (BreathQuest, VaakMirror, Chime, VoiceHurdleRace) across 8 Indian languages.
- Implemented phoneme-level acoustic verification evaluating place, manner, and voicing errors with sub-300ms audio latency and automated clinician report generation.
- Real-World Impact:** Deployed clinical assessment system actively used for objective pediatric speech screening across 8 Indian languages.

Vaakify — Pediatric Multilingual Speech Practice Platform

[Live Platform](#) · [GitHub: Speech-Repeater](#)

Python, Librosa, Praat-Parselmouth, Faster-Whisper, Coqui TTS, FastAPI, React

- Engineered a phoneme-level articulation engine extracting jitter, shimmer, formant frequencies (F1/F2), and pitch contours from speech, powering clinical practice in English, Hindi, and Kannada.
- Integrated automated phonetic transcription via Epitran G2P to measure distance metrics against target speech and generate targeted phoneme drills.
- Real-World Impact:** Provides objective, quantitative articulatory biofeedback directly to pediatric speech therapists, replacing subjective manual error scoring.

NIRVANA (Firebot) — Autonomous Firefighting Robot & MuJoCo Simulation

[Live Render Console](#) · [GitHub: Firebot](#)

Python, MuJoCo 3D Physics, PyTorch, Gymnasium, EIF Sensor Fusion, FastAPI, WebSockets

- Engineered an autonomous firefighting robot in a physics-backed MuJoCo 3D simulator across procedural multi-room environments with volumetric particle fire modeling.
- Implemented MM-FusionRL (multimodal cross-attention deep RL) coupled with Extended Information Filter (EIF) sensor fusion for thermal localization and obstacle navigation.
- **Real-World Impact:** Deployed live simulation console on Render eliminating hazardous human entry in structural fires by localizing heat sources in <45s inside physics environments.

Tripwire — Real-Time CPU-Only Edge CV Vehicle Counter

[Live Demo](#) · [GitHub: Vehicle-Counter](#)

Python, OpenCV, FastAPI, React, Vite, WebSockets, Docker

- Engineered an edge-ready, GPU-free computer vision vehicle detection and tracking pipeline from live CCTV feeds operating with sub-50ms latency.
- Implemented Gaussian background subtraction, morphological noise filtering, centroid tracking, and directional virtual tripwire counting streamed over WebSockets.
- **Real-World Impact:** Cuts camera surveillance hardware costs by 80% by delivering accurate multi-lane traffic tracking on standard commodity CPUs without dedicated GPUs.

GECO-Enhanced — Confidence-Aware Semantic Keypoint Correspondence

[GitHub: GECO](#)

PyTorch, DINOv2 (ViT), Python Optimal Transport (POT), FastAPI, React, Docker

- Formulated an unsupervised semantic correspondence framework establishing keypoints across different instances of identical object classes under severe pose and perspective variation.
- Extracted dense DINOv2 ViT features and solved correspondence using Sinkhorn-regularized Optimal Transport with confidence weighting and entropy regularization.
- **Real-World Impact:** Enables zero-shot visual alignment across varying camera viewpoints without requiring expensive annotated paired training datasets.

EDUCATION & CERTIFICATIONS

- **Bachelor of Technology (B.Tech)**, SASTRA University (Expected 2027)
- **SWAYAM – IIT Madras:** Comprehensive Electronics and Embedded Systems (*In-Person Practical Program*)